Appendix A: Data Description

A.1 Data Sources

Quarterly Performance Reports (QPR). The QPR data is extracted from HUD's DRGR (Disaster Recovery Grant Reporting) system. This administrative data captures:

- Fund allocations by appropriation
- Obligation, disbursement, and expenditure amounts
- Activity types (housing, infrastructure, economic development, planning, administration)
- Quarterly reporting periods (2003-2023)

The raw dataset contains 130,605 quarterly observations across 78 unique grantees and 206 grant allocations.

Population Data. Population figures are obtained from:

- U.S. Census Bureau decennial census (2000, 2010, 2020)
- American Community Survey estimates for intercensal years

Disaster Characteristics. Disaster type classifications derive from:

- FEMA declared disaster data

- Primary hazard type (hurricane, wildfire, flood, other)
- Disaster declaration date for era classification

Experience Indicators. Grantee experience measures are computed from:

- Count of prior CDBG-DR grants managed
- Binary classification: Novice (first CDBG-DR grant) vs. Experienced (2+ prior grants)

A.2 Spending Velocity Construction

The Dynamic Denominator Problem. Initial velocity calculations using period-specific denominators produced computational artifacts. When the denominator (obligated amount) changes across periods, apparent velocity changes reflect administrative adjustments rather than actual spending acceleration:

$$\text{Velocity}_t^{\text{raw}} = \frac{\text{Expended}_t}{\text{Obligated}_t} - \frac{\text{Expended}_{t-1}}{\text{Obligated}_{t-1}}$$

This formula produces spurious velocity when obligations change. For example, if Obligated doubles from quarter $t-1$ to t while Expended increases moderately, the ratio drops despite actual spending increases.

Standardized Velocity with Fixed Denominators. The standardized approach uses the final obligated amount as a stable denominator:

$$\text{Velocity}_t^{\text{std}} = \frac{\text{Expended}_t}{\text{Obligated}_{\text{final}}} - \frac{\text{Expended}_{t-1}}{\text{Obligated}_{\text{final}}}$$

This ensures that velocity changes reflect only changes in the numerator (actual expenditure), not administrative adjustments to obligations.

Impact of Standardization.

Metric	Raw Velocity	Standardized Velocity	Reduction
Std Dev	0.089	0.028	68.5%
Extreme values (>	0.10	)	0.60%
IQR	0.031	0.012	61.3%

The standardized approach dramatically reduces measurement noise while preserving meaningful velocity variation.

Winsorization. Standardized velocity values are winsorized at the 1st and 99th percentiles to limit outlier influence:

$$\text{Velocity}^{\text{winsorized}} = \begin{cases} P_1 & \text{if Velocity} < P_1 \\ P_{99} & \text{if Velocity} > P_{99} \\ \text{Velocity} & \text{otherwise} \end{cases}$$

This bounds extreme observations while preserving the full distribution shape.

A.3 Phase-Specific Velocity

Programs are divided into temporal terciles based on elapsed quarters:

- **Early phase**: Quarters 1 through $\lfloor T/3 \rfloor$
- **Middle phase**: Quarters $\lfloor T/3 \rfloor + 1$ through $\lfloor 2T/3 \rfloor$
- **Late phase**: Quarters $\lfloor 2T/3 \rfloor + 1$ through T

where T is the total number of observed quarters.

Mean velocity within each phase creates phase-specific measures:

$$\text{Velocity}_{\text{phase}} = \frac{1}{N_{\text{phase}}} \sum_{t \in \text{phase}} \text{Velocity}_t$$

A.4 Sample Characteristics

Overall Sample Composition.

Characteristic	**Count**	**Percent**
Total Observations	152	100.0%
Unique grantees	78	–
Unique disaster events	18	–
Unique grant allocations	206	–

Government Type Distribution.

Government Type	**N**	**Percent**
State governments	37	24.3%
Local governments	40	26.3%

Government Type	**N**	**Percent**
Multiple/Mixed	75	49.3%
Total	**152**	**100.0%**

Disaster Type Distribution.

Disaster Type	**N**	**Percent**
Hurricane	89	58.6%
Flood	27	17.8%
Wildfire	24	15.8%
Other	12	7.9%
Total	**152**	**100.0%**

Experience Distribution.

Experience Level	**N**	**Percent**
Novice (first CDBG-DR)	74	48.7%
Experienced (2+ prior)	78	51.3%
Total	**152**	**100.0%**

Program Type Distribution.

Primary Program Type	N	Percent
Housing	67	44.1%
Infrastructure	41	27.0%
Administration	34	22.4%
Other	10	6.6%
Total	**152**	**100.0%**

Censoring Status.

Status	N	Percent
Completed (reached 95% expenditure)	40	26.3%
Right-censored (not yet complete)	112	73.7%
Total	**152**	**100.0%**

The high censoring rate (73.7%) motivates survival analysis methods. Standard regression approaches that drop censored observations would discard 112 of 152 observations, severely limiting statistical power and introducing selection bias.

A.5 Velocity Summary Statistics

Variable	N	Mean	SD	Min	Median	Max
Overall Velocity (pp/quarter)	152	2.14	1.83	0.12	1.67	9.84
Early Phase Velocity	152	2.89	2.41	0.08	2.21	12.67
Middle Phase Velocity	152	1.92	1.76	0.04	1.48	8.93
Late Phase Velocity	152	1.61	1.89	0.02	1.12	11.24
Duration (quarters, completed only)	40	68.2	28.4	18	63.5	142

Key observations:

1. **Overall velocity** averages 2.14 percentage points per quarter, indicating that the typical grantee expends approximately 2% of their total allocation each quarter.

2. **Phase pattern**: Velocity is highest in early phases (2.89 pp/quarter) and declines through middle (1.92) and late (1.61) phases, consistent with front-loaded activity followed by closeout delays.

3. **Substantial variation** exists across grantees (SD = 1.83 for overall velocity), providing analytic leverage for detecting velocity-completion associations.

4. **Completion timing**: Among the 40 completed programs, median time to 95% completion is 63.5 quarters (approximately 16 years).

A.6 Data Quality

Quality assurance flags track potential measurement issues:

QA Flag	N	Percent
Extreme velocity (	v	> 10 pp/quarter)
Large obligation adjustment (>25%)	12	7.9%
Negative expenditure adjustment	8	5.3%

Sensitivity analyses excluding flagged observations produce substantively similar results (see Appendix C).

A.7 Disaster Coverage

The sample covers 18 major disaster events from 2003 to 2023:

Hurricane Events: - Hurricane Katrina (2005): Gulf Coast states - Superstorm Sandy (2012): Mid-Atlantic and Northeast - Hurricane Harvey (2017): Texas coast - Hurricane Maria (2017): Puerto Rico and U.S. Virgin Islands - Hurricane Irma (2017): Florida and Caribbean

Wildfire Events: - California Wildfires (2017-2020): Multiple events - Colorado Fires (2012-2013) - Arizona Wildfires (2011)

Flood Events: - Louisiana Floods (2016) - Colorado Floods (2013) - Multiple smaller flood events

Other Events: - Joplin Tornado (2011) - Oklahoma Tornadoes (2013) - Various smaller disasters

The diversity of disaster types, geographic locations, and governmental structures provides variation needed to test context-specific velocity effects.